

Phase 0 — Execution Runbook & Pre-Gate Prep

LocalAI TV · v1.3-aligned · Prep While the Validation Gate Is Held

To: Sheikh Sameer & Gnana Rajnan (execution) · Nagarjuna Reddy (CTO) · Koneti Mohan Reddy (Founder) **From:** LocalAI TV Architecture Team **Date:** 16 May 2026 **Status:** 🟡 **PREP DOCUMENT — execution is GATED.** The Validation Gate is intentionally held. Only **Part A (pre-gate prep)** is actionable now. **Part B (execution) starts only when the Founder explicitly lifts the gate.**

Rule (unchanged): No EC2 launch, no pipeline change, no Phase 1, no coding past Phase 0 until the Founder says "start." This runbook is written *now* so that the moment the gate lifts, Sameer & Gnana execute Part B with zero scramble. Nothing in Part B is to be run before that.

Part A — Pre-Gate Action Items (DO NOW, in parallel — no gate crossed)

These four items involve **no pipeline change and no Phase 0 execution**. They are pure readiness and remove start-day delays.

A0 — Request the AWS GPU quota increase (highest lead-time item — do first)

AWS blocks GPU instances by default. The account almost certainly has a **0 limit** for the instance family Phase 0 needs.

- **Where:** AWS Console → **Service Quotas** → **Amazon EC2** → search "**Running On-Demand G and VT instances**" → Request quota increase.
- **Region:** **Asia Pacific (Mumbai)** **ap-south-1** (must match — quotas are per-region).
- **Request:** enough vCPUs for **g4dn.xlarge** (4 vCPUs). Request **at least 8** to allow one running + one test instance.
- **Lead time:** AWS approval typically **1–2 business days**. This is the single biggest risk to a fast Phase 0 start — **submit today**.
- **Owner:** Nagarjuna / DevOps. **No cost** to request.
- ☐ Submitted ☐ Approved (record date)

A1 — Send the 3 non-secret readiness confirmations

Plain text to the Architecture Team — **descriptions only, no passwords/keys**. Lets us verify Phase-0 readiness and flag gaps now.

Item	Confirm
AWS	Account ready? Region <code>ap-south-1</code> (Mumbai)? Same account as live channels (Y/N)? GPU quota (A0) requested?
Postgres	Version 15? Running where (Hostinger VPS / other)? Reachable from where the pipeline runs?
Domain / DNS	<code>localaitv.com</code> DNS managed at which registrar? Is <code>content.localaitv.com</code> free to point at the CDN?
VPS	Hostinger — which OS/version? Pipeline currently runs here?

- **Owner:** Nagarjuna. ☐ Sent

A2 — Stage credentials in the shared vault (NOT to the Architecture Team / not in chat)

- Create the shared vault (Bitwarden free is fine) → invite **only Sameer & Gnana**.
- Put in: AWS IAM access key (least-privilege, see A3), VPS SSH, Postgres connection string.
- **Never** send these to this assistant, chat, email, or WhatsApp.
- **Owner:** Founder / Nagarjuna. ☐ Vault ready ☐ Sameer & Gnana invited

A3 — Guard-rails on the existing AWS account (since it also runs live channels)

So Phase 0 testing **cannot touch the live service**:

- ☐ **Dedicated IAM user** for Sameer/Gnana — least privilege: only the Phase 0 S3 bucket + launch/stop the Phase 0 EC2 instance. **Not** root, **not** full access.
- ☐ **Separate S3 bucket** `localaitv-content-mumbai` (distinct from any live-channel bucket).
- ☐ **Billing budget alert** (e.g. ₹35,000/mo) so test spend is visible & isolated.
- ☐ **Tag** every Phase 0 resource `project=phase0` so it's never confused with live resources.
- **Owner:** Nagarjuna / DevOps.

Part A creates nothing in production and crosses no gate. It is the correct use of the hold period.

Part B — Phase 0 Execution Runbook (RUN ONLY AFTER THE FOUNDER LIFTS THE GATE)

Scope: Two parallel tasks, run against the **existing Gupshup / direct-API flow**. **No webhook**. **No Admin Dashboard**. **No pipeline logic change** except one FFmpeg flag.

Step	Action	Owner
B1	Launch EC2 g4dn.xlarge in <code>ap-south-1</code> , tagged <code>project=phase0</code>	Gnana
B2	Install: CUDA 12.x + NVIDIA driver 535+ + FFmpeg built with <code>--enable-nvenc</code> + Python 3.11	Gnana
B3	<code>git clone</code> the existing pipeline code — no code changes	Gnana
B4	Apply the single FFmpeg flag change (Part C) — nothing else in build logic	Gnana
B5	Verify S3 read/write from the EC2 instance to <code>localaitv-content-mumbai</code>	Sameer
B6	Wire <code>s3_storage.py</code> upload into <code>run_planner_task</code> after <code>build_bulletin_video()</code> succeeds (one post-success call — touch nothing else)	Sameer
B7	Run one full bulletin end-to-end through the existing flow	Sameer + Gnana
B8	Record G1–G4 numbers in Part D; hand the filled table to the Founder	Sameer + Gnana

If any Phase 0 task starts needing a webhook, HMAC, or the Admin Dashboard — **stop**: it is mis-scoped. Phase 0 has no upstream dependency.

Part C — Config Templates (no secrets — real values come from the vault)

The **ONLY** pipeline change in Phase 0 (FFmpeg encoder swap):

```
# In the FFmpeg command(s):
- -c:v libx264 -preset fast
+ -c:v h264_nvenc -preset p4 -tune hq
```

Environment template (`.env` — placeholders only; real values from the vault):

```
AWS_REGION=ap-south-1
S3_BUCKET=localaitv-content-mumbai
S3_PUBLIC_CDN_BASE_URL=https://content.localaitv.com
AWS_ACCESS_KEY_ID=<from vault>
AWS_SECRET_ACCESS_KEY=<from vault>
# Output key convention:
# ai-processed/<state>/<district>/<constituency>/<category>/<content_id>/<asset_role>.<ext>
```

No HMAC secret / callback token in Phase 0 — those are Phase 1 (webhook), which is gated.

Part D — Gate Results Capture (Sameer & Gnana fill this; Founder reviews to decide pass/fail)

Criterion	Target	Measured	Pass?
G1 — Encoding latency	Bulletin build < 2 min (target ~90 s) vs. ~15 min baseline	Baseline: ____ · NVENC: ____	☐
G2 — S3 upload reliability	100% of test builds land at the correct <code>s3://localaitv-content-mumbai/ai-processed/...</code> key	____ / ____ builds	☐
G3 — CDN delivery	Same media fetchable via <code>https://content.localaitv.com/ai-processed/...</code> (not the S3-direct URL)	____	☐
G4 — End-to-end availability	Media playable through the CDN within the build cycle, no manual steps	____	☐

Gate verdict: ☐ PASS (all four) → Founder reviews → Founder says "start" → Phase 1 unlocks · ☐ FAIL → fix & re-test, stay in Phase 0.

Part E — What "Gate Cleared" Unlocks (for awareness — not now)

Only after G1–G4 pass **and** the Founder explicitly says "start":

- Scope freeze locked (Gnana + Abishek sign-offs in)
- Plan **v1.4** issued in one pass (Addendum A items 1–11)
- Phase 1 build begins (Admin Dashboard + webhook adapter)

Until then: **gate held, Part A only.**

Sign-off

Party	Action	Status
Nagarjuna (CTO)	Run Part A (A0 quota, A1 confirmations, A2 vault, A3 guard-rails)	☐
Sameer & Gnana	Stand by for Part B; review runbook, flag issues	☐
Founder — Koneti Mohan Reddy	Lifts the gate when scope frozen + ready	☐

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Prep only. Part B does not run until the Founder lifts the gate. Plan v1.3 unchanged.